



Additive Manufacturing of Lunar Regolith: A Review

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ABSTRACT

Lunar in-situ construction using additive manufacturing (AM) technology has emerged as a critical pathway for sustainable extraterrestrial exploration. This review systematically evaluates two dominant AM paradigms for lunar regolith processing: low-temperature deposition forming (material extrusion and binder jetting), and high-energy beam additive manufacturing (powder bed fusion and directed energy deposition). Low-temperature methods achieve moderate compressive strength with low energy consumption but face challenges such as binder dependency and vacuum instability. By contrast, high-energy beam techniques enable binder-free fabrication with better compatibility for in-situ resource utilization, though they suffer from porosity, high energy intensity, and geometric limitations. In the context of lunar in-situ resource utilization (ISRU), low-temperature methods offer near-term feasibility for small-scale infrastructure, while high-energy approaches show promise for large-scale, autonomous construction by leveraging solar energy and raw regolith. Future advancements will hinge on hybrid systems that integrate material efficiency, energy sustainability, and robotic adaptability to overcome extreme environmental challenges. This review consolidates technological progress, identifies interdisciplinary synergies, and provides strategic insights into guiding the transition from Earth-dependent prototypes to self-sufficient lunar habitats, ultimately advancing the capability of humanity for a long-term extraterrestrial presence.

1. Introduction

Space exploration, a core domain in the advancement of human science and technology, epitomizes humanity's relentless pursuit of expanding its cognitive boundaries. Recent advancements in aerospace technology have accelerated global efforts in deep-space exploration, with extraterrestrial exploration and construction emerging as key areas of international focus. Research in extraterrestrial construction spans a range of disciplines, including planetary science, materials science and engineering, mechanical engineering, civil engineering, and geological engineering, highlighting its inherently interdisciplinary nature.

As the Earth's closest celestial neighbor, the Moon serves not only as a critical stepping stone for deep-space exploration, but also as a vital platform for future scientific research and technological experimentation owing to its unique environmental conditions and resource potential. Against this backdrop, nations worldwide have formulated and implemented lunar exploration and development programs, encompassing unmanned missions, crewed landings, resource extraction, and lunar base construction [1–5]. Among these initiatives, in-situ lunar construc-

tion, particularly the establishment of permanent habitats and infrastructure [6,7], has become a pivotal focus for advancing sustainable lunar exploration [8–13].

Extraterrestrial construction methodologies can be broadly categorized into four types: prefabrication–launch–landing, prefabrication–launch–deployment, prefabrication–launch–assembly, and in-situ extraterrestrial construction. While the first three approaches rely on earth-based prefabrication and subsequent transportation of components for on-site assembly or deployment, in-situ construction emphasizes utilizing local extra-terrestrial resources to minimize dependence on Earth-bound logistics. This paradigm shift is driven by the limitations of current Earth–Moon transportation capabilities and costs as well as the need to develop scalable solutions for future missions to Mars and beyond. Central to this strategy is in-situ resource utilization (ISRU), which aims to harness lunar materials to produce life-support consumables, energy, propellants, and construction materials, thereby enabling cost-effective, efficient, and sustainable exploration and habitation [14–17].

The lunar regolith, a loose granular layer covering the surface of the Moon, is a primary target of ISRU research [18–20]. Formed through bil-

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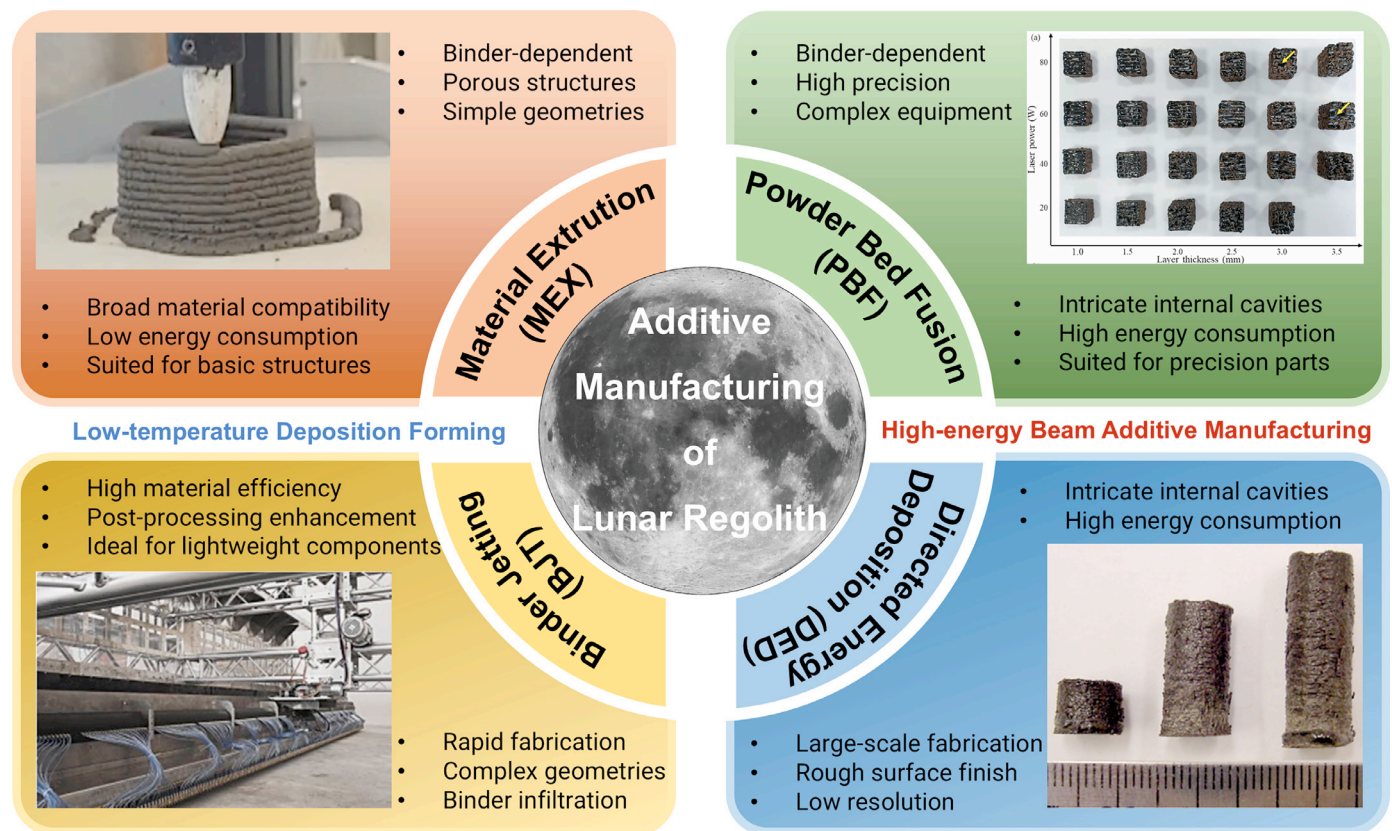


Fig. 1. Schematic of additive manufacturing technologies of lunar regolith simulants.

lions of years of meteoritic bombardment and solar-wind interactions, this heterogeneous material comprises oxides, silicates, and other mineral phases that exhibit unique physicochemical properties. As a composite of silicates and glassy particles, lunar regolith presents unique challenges for AM, including an irregular particle morphology, high melting temperatures, and vacuum compatibility. Research on various methods of extraterrestrial construction [21–23] has led to significant advancements in additive manufacturing (AM) technologies, revolutionizing the potential of lunar regolith as a construction material [24–28]. By adapting 3D printing techniques, researchers aim to fabricate robust structures capable of withstanding extreme temperature fluctuations, radiation, and micrometeoroid impacts—factors critical for long-term lunar habitation [29–36].

According to the ISO/ASTM 52900 classification standard [37], AM technologies can be divided into seven categories: binder jetting (BJT), directed energy deposition (DED), material extrusion (MEX), material jetting (MJT), powder bed fusion (PBF), sheet lamination (SHL), and vat photopolymerization (VPP). In the context of 3D printing lunar regolith simulants, the primary AM technologies involve MEX, BJT, PBF, and DED, which can be classified as low-temperature extrusion forming or high-energy beam additive manufacturing (Fig. 1).

This review systematically examines state-of-the-art lunar regolith-based additive manufacturing, with a focus on the aforementioned dominant technological paradigms. By analyzing their principles, advancements, and limitations, this review aims to provide a comparative assessment of energy efficiency, structural performance, and in-situ compatibility. It seeks to identify critical challenges in material-process-environment interactions under lunar conditions and outline future research directions to bridge the gap between laboratory-scale demonstrations and mission-ready technologies. The insights presented herein will inform the design of next-generation lunar construction systems, ultimately supporting the ambition of humanity for a sustained presence on the Moon and beyond.

2. Low-Temperature Deposition Forming

2.1. Material extrusion (MEX)

MEX involves the deposition of a concrete-like slurry composed of an extra-terrestrial regolith powder that solidifies into structural components through subsequent drying and curing processes. In lunar regolith additive manufacturing, two key methods are employed: direct ink writing (DIW) for precision deposition and fused deposition modeling (FDM) using thermoplastic-regolith composites.

2.1.1. Direct ink writing (DIW)

Research on lunar regolith utilization via DIW has primarily focused on three material systems: biopolymer binding, matrix transformation, alkali activation, and polymers with curing or sintering, each demonstrating unique merits and limitations in terms of printability, mechanical performance, and environmental adaptability.

Biopolymer-bound soil composites (BSCs) utilizing proteins (e.g., bovine blood, human serum albumin) achieve a compressive strength of 39.7 MPa through urea-induced β -sheet networks. These composites offer recyclability and 3D-printability, despite moisture-dependent curing limitations [38,39] (Fig. 2(a)). Matrix transformation methods (the solidification process involves the formation of a crosslinked matrix holding together individual particulates of the soil rather than direct dissolution and reconfiguration of aluminosilicate frameworks) enable low-energy regolith stabilization (<5 % energy of sintering), although restricted to silica-rich mare soils [40] (Fig. 2(b)). Alkali-activated geopolymers (AAGs) formed by geopolymerization (aluminosilicates form a three-dimensional network upon activation under typically alkaline conditions with the complete dissolution of aluminosilicates into monomers, dimers, and trimers, followed by the rapid hardening of the produced slurry through the formation of a 3D aluminosilicate network) achieve 48.1 a compressive strength under lu-

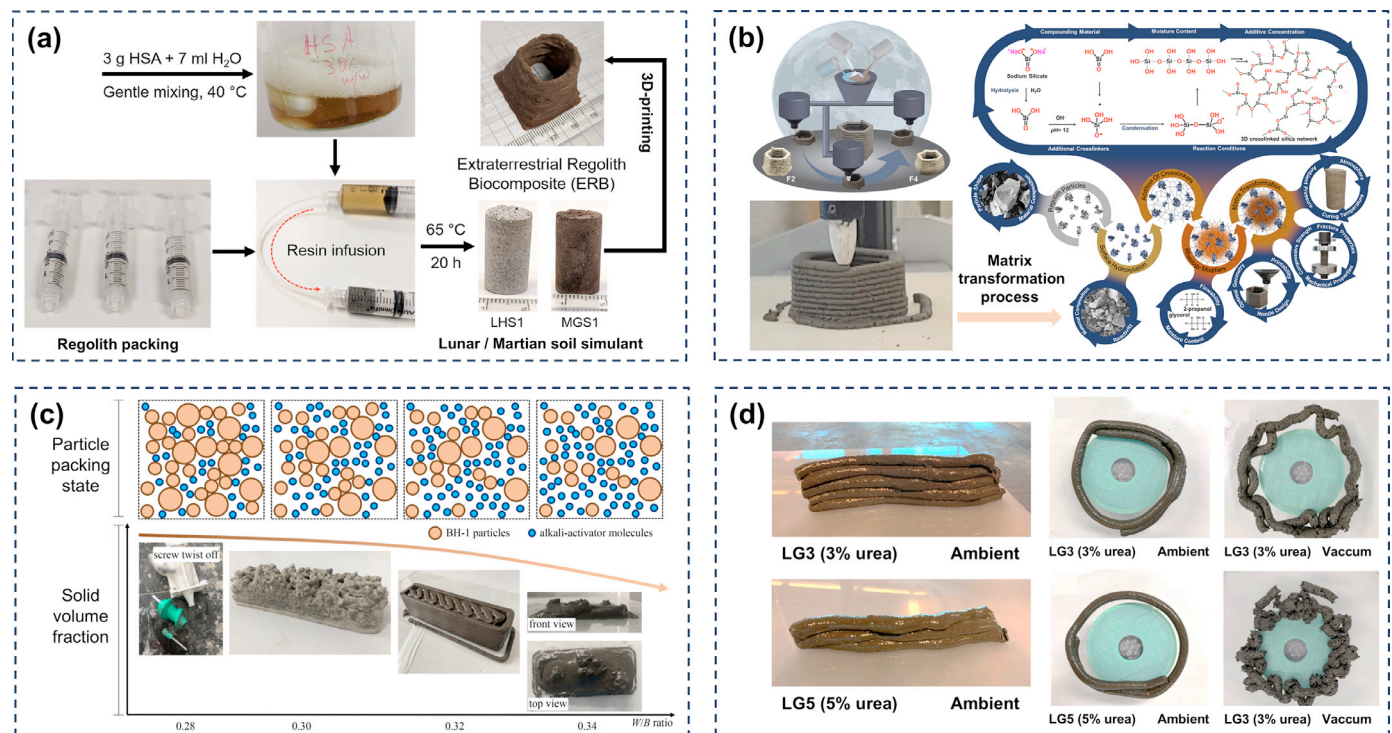


Fig. 2. DIW fabrication of (a) extraterrestrial regolith bio-composites [38], (b) matrix transformed lunar regolith showing the reaction scheme for surface dissolution, functionalization, and condensation yielding mineral phases enmeshed in a crosslinked siloxane network incorporating additional multifunctional molecules, (c) alkali-activated geopolymers, and (d) geopolymers with different urea dosages [46,47].

nar conditions but faced anisotropic layer bonding challenges [41–45] (Fig. 2(c)). The viscosity of the pastes decreased by approximately one order of magnitude when the urea concentration increased from 3 wt % to 5 wt %. The addition of urea reduced the porosity and improved the vacuum/high-temperature stability, while lowering the yield stress [46,47] (Fig. 2(d)). These approaches collectively address critical aspects of lunar construction: material efficiency, energy minimization, structural integrity, and process adaptability; however, they require further optimization for vacuum-phase stability and interlayer cohesion.

In lunar regolith simulant-based geopolymer composites, bioinspired 3D printing strategies integrate carbon fiber/quartz sand reinforcements into geometric designs (honeycomb, triangular, wave, and rectangular, as shown in Fig. 3) [48]. All bio-inspired structures exhibited anisotropic mechanical properties. Along the Y-axis, the rectangular structures exhibited the highest compressive strength (20.1 MPa), followed by the triangular (17.9 MPa), honeycomb (15.6 MPa), and wavy (11.3 MPa) designs. Z-axis testing revealed superior performance for honeycomb patterns (46.7 MPa), exceeding that of the rectangular (34.4 MPa), triangular (26.5 MPa), and wavy (23.8 MPa) variants. The fracture strains ranged from 5.8 % to 10.2 % (Y-axis) and 12.1 % to 13.9 % (Z-axis), with all structures exhibiting non-catastrophic failure modes and notable impact resistance. The compressive strength in the Z-axis direction was significantly higher than that in the Y-axis direction, significantly outperforming most other 3D-printed cement- and geopolymer-based materials with various architectures.

In most polymer–regolith systems, lunar regolith relies on binder systems that ensure rheological stability, rapid solidification under extraterrestrial conditions, and compatibility with post-processing methods such as curing or sintering. The UV/thermal dual-curable resin achieved 86 wt % regolith loading, producing structures with 9.83 MPa compressive strength via sequential curing [49]. However, acrylate binders degrade under lunar thermal cycling (−190 °C to 127 °C), increasing porosity and surface erosion [50,51]. The PLGA-LRS composites (90 wt % regolith) formed porous architectures (27.9 % porosity) (Fig. 4(a)) with elastic recovery below 20 % strain (Fig. 4(b)). Sintering PLGA-LRS green

bodies at 1100 °C yielded ceramics with 47 % density and 8.4 MPa strength in air, or ceramics with 38 % density under hydrogen (Fig. 4(c)) [52]. While enabling rapid prototyping, PLGA-based DIW faces mechanical constraints and high energy demands for extraterrestrial applications. Moreover, hydrogel–regolith composites, integrating hydrophilic polymers with a lunar simulant and water, have been explored for extrusion-based additive manufacturing. When the regolith content was increased from 25 to 40 vol % reduced sintered porosity (6.4 % to 2.18 %) and enhanced density (0.78 to 0.89 g/cm³) and compressive strength (0.68 to 1.32 MPa) [53,54]. These materials demonstrate printability for complex geometries, validated through prototype tools and structural blocks [55], and aligned with lunar infrastructure requirements.

2.1.2. Fused deposition modeling (FDM)

Fused deposition modeling (FDM), also known as fused filament fabrication (FFF), is a widely adopted AM technique that constructs objects by extruding thermoplastic filaments through heated nozzles. The FDM of lunar regolith–polymer composites has garnered significant attention for in-situ manufacturing applications, where the molten sediment can be made of thermoplastic or sulfur concrete. Thermoplastic composites (ABS, nylon, and PEEK) exhibit trade-offs: regolith reinforcement enhances strength but reduces toughness, requiring high extrusion temperatures (e.g., 235 °C for ABS) [56] or pressure-optimized interfaces (e.g., 36.2 MPa tensile strength for nylon) [57]. The PEEK–regolith composites exhibited a 26.8 % strength reduction at 30 wt % owing to porosity [58]. PLA produced vacuum-compatible 80:20 regolith blends with recyclability [59,60] (Fig. 5). Concurrent engineering developments include the integration of FDM-printed conical centrifuges for regolith particle classification, with an emphasis on surface roughness modulation as a critical parameter for efficient in-situ resource utilization [61].

2.1.3. Sulfur–regolith concrete (SRC)

From the experiments shown in Fig. 6, sulfur–regolith concrete (SRC) outperformed thermoplastics with rapid curing, achieving a peak com-

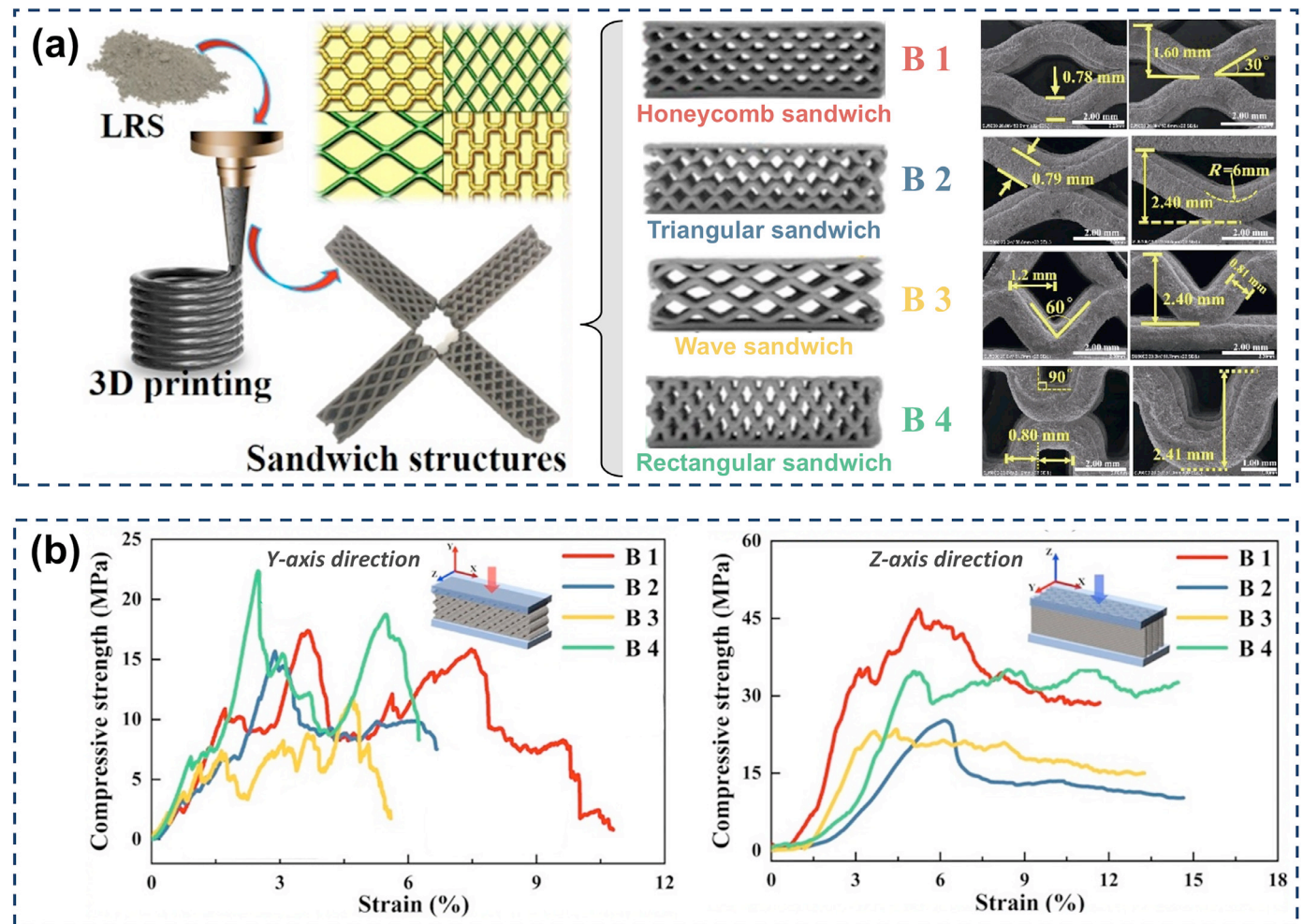


Fig. 3. Regolith–geopolymer composites with bioinspired sandwich structures of (a) honeycomb, triangular, wave, rectangular shapes, and (b) corresponding compressive strength measured in different directions [48].

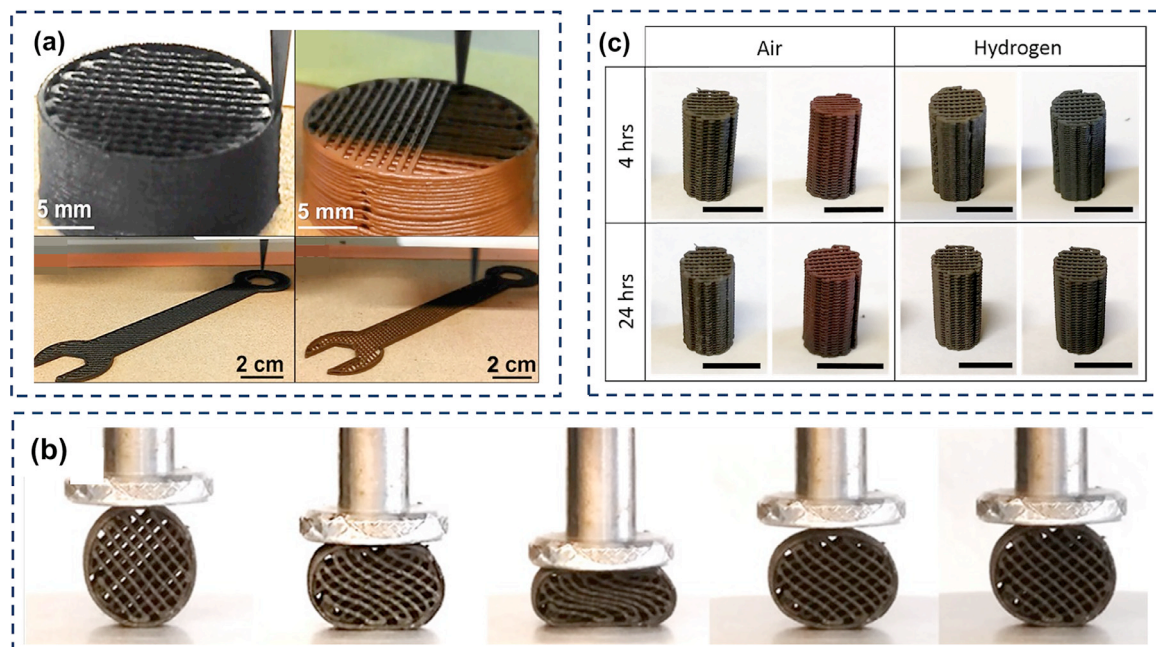


Fig. 4. (a) 3D-printed multilayered PLGA–regolith composite cylinder, (b) cylinder undergoing a single compression cycle, and (c) printed samples undergoing sintering under different conditions [52].

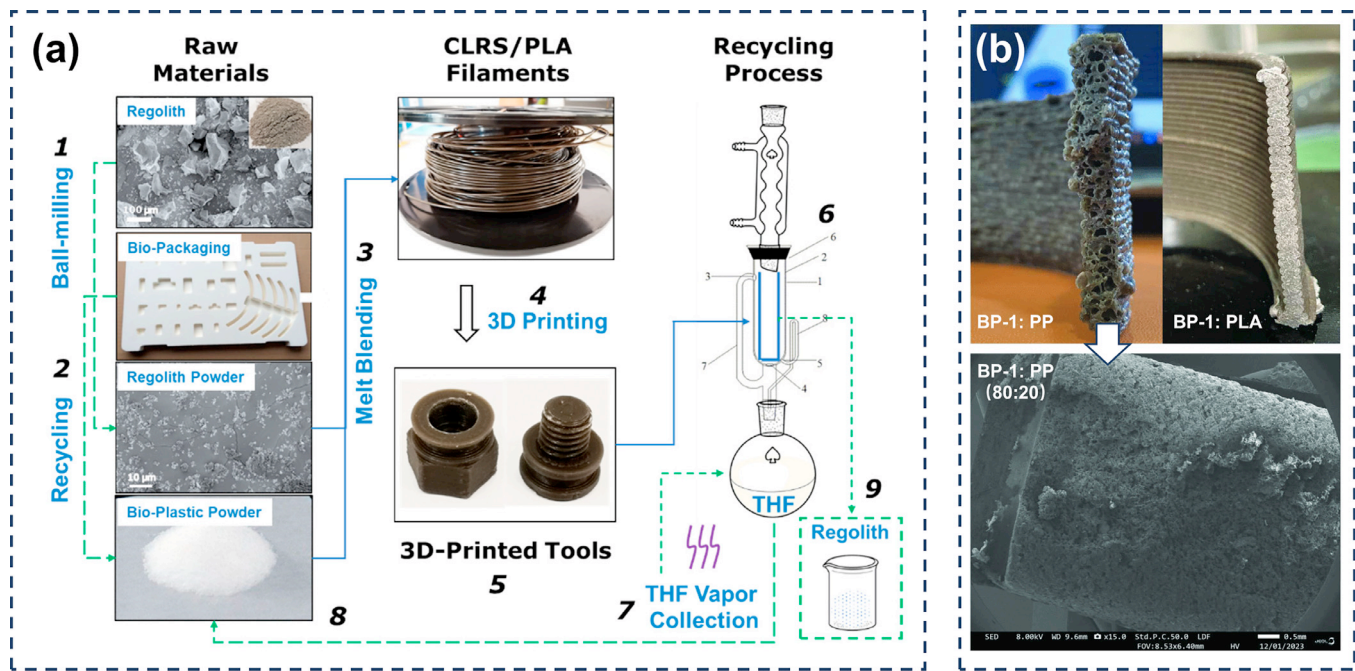


Fig. 5. (a) Fabrication and recycling of the regolith-polymer composite (RPC), and (b) observation of the lunar regolith simulant-polymer composite of BP-1: PP and BP-1: PLA [59,60].

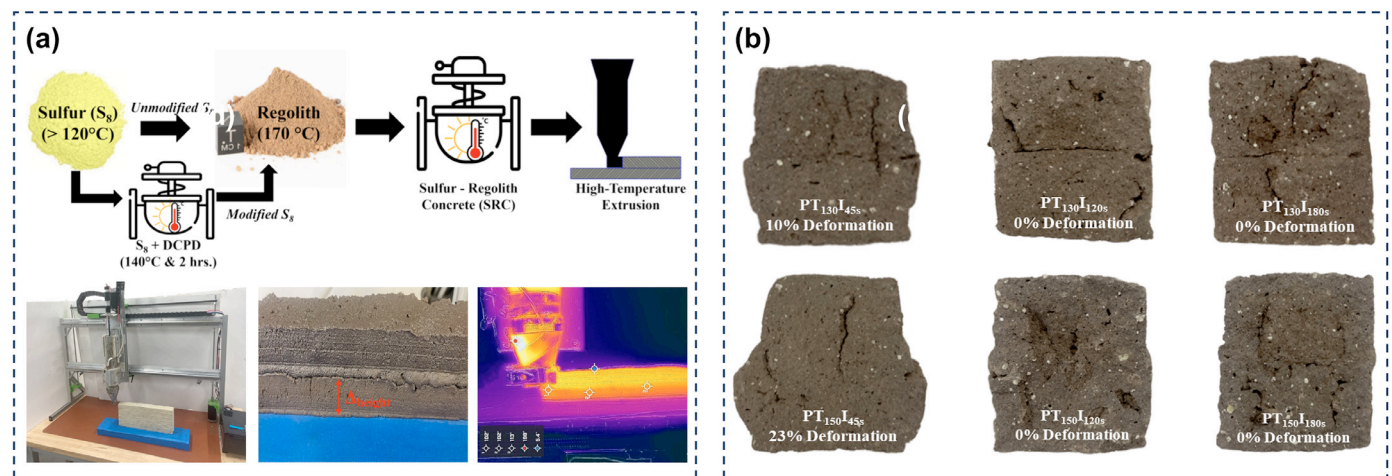


Fig. 6. (a) SRC preparation and testing process, and (b) cross-sectional profiles of printed SRC specimens post-shape-stability-test [62].

pressive strength at 30 wt % sulfur (100–200 % UCS increase under 1–2 MPa compaction) [62,63], though vacuum exposure causes 9 %–12 % strength loss [64,65]. Critical challenges include sulfate formation from lunar oxides, interlayer adhesion degradation (22 % decrease in flexural strength with delayed printing), and environmental compatibility. These findings highlight the material-process-property interdependencies of extraterrestrial manufacturing.

2.2. Binder jetting (BJT)

The BJT, initially developed in 1989 by the MIT team, selectively deposits liquid binders onto layered powders to fabricate solid structures. Unbound powder is removed post-printing using air jets, offering advantages in resolution and scalability [66,67]. This method is particularly notable for its scalability and high resolution, which enable complex geometries that are unattainable through conventional methods. Magnesium oxychloride (MOC)-regolith composites have demonstrated potential for use in the lunar infrastructure. Sintering param-

eter optimization for regolith simulants revealed temperature-dependent densification behavior [68]. The in-situ fabrication of functional tools via BJT has been verified, although challenges persist regarding material anisotropy and interlayer adhesion caused by binder-powder interaction dynamics.

The d-shaped method, a pivotal BJT technology developed under ESA funding, enables large-scale lunar construction by depositing regolith-binding ink through a gantry-mounted printhead that traverses the X- and Y-axis (Fig. 7).

The d-shaped additive manufacturing process utilizes a layer-by-layer deposition method to bind granular materials such as lunar regolith simulants into cohesive structures [69]. This approach employs a liquid-binding agent selectively deposited through a multi-nozzle system, enabling precise control over the material distribution and structural geometry, highlighting its great potential for resource-efficient, autonomous extraterrestrial habitat construction. With the integration of CCD cameras, real-time process monitoring can enhance the accuracy by comparing printed layers with digital models [70].

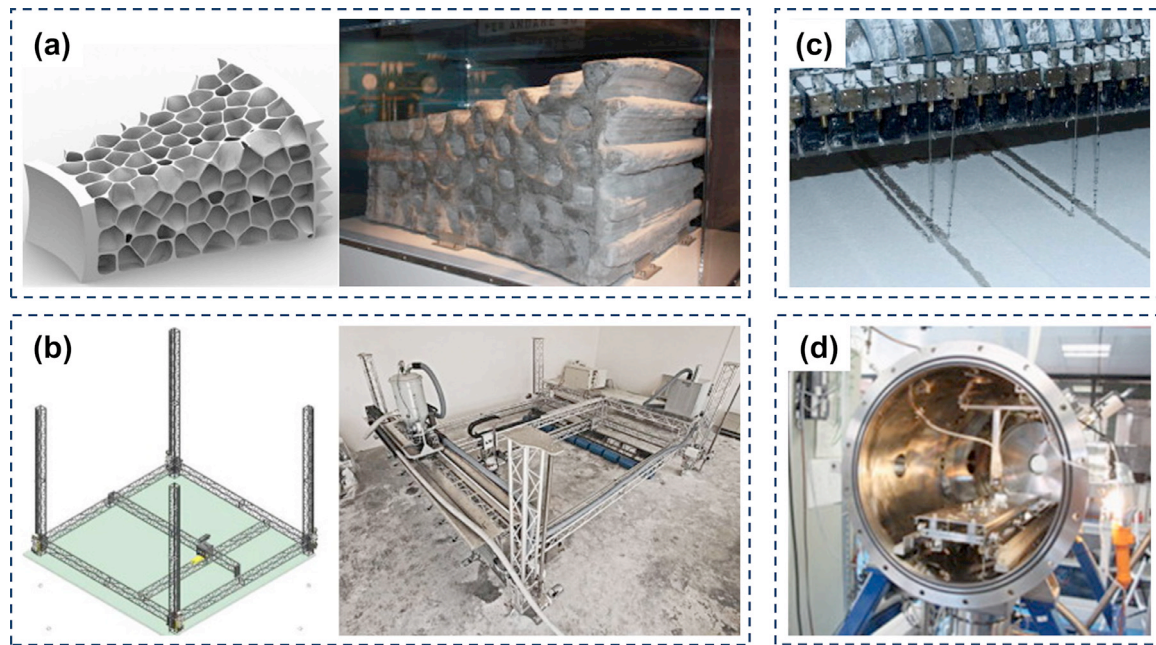


Fig. 7. d-shape method [69]: (a) Exemplar structural element; (b) Material depositing unit and the entire printer; (c) Printing head with 300 nozzles and the printing process, and (d) Vacuum chamber and ink feeding-through system.

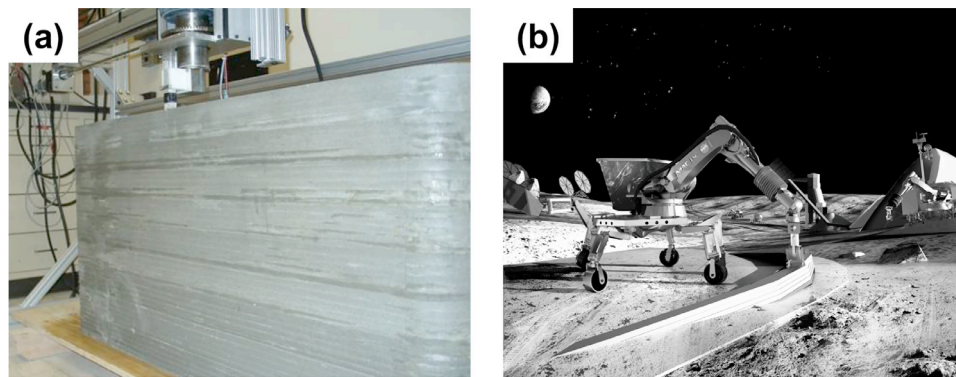


Fig. 8. Contour-crafting technology operation (a) on earth [72] and (b) lunar hosting by an ATHLETE rover [73].

2.3. Contour crafting (CC)

Contour crafting (CC) technologies play an important role in 3D printing and continue to create breakthrough advancements (from Contour Crafting, Inc.). CC technologies are additive manufacturing methods based on robot extrusion and troweling technology, and show significant potential in the field of extraterrestrial construction [71] (Fig. 8). These technologies use computer-controlled nozzles to deposit high-viscosity materials (such as concrete or lunar soil mixtures) layer by layer, enabling template-free and rapid construction of complex structures. They support embedded pipelines and other infrastructure, greatly reducing dependence on Earth's resources and human labor. Based on their comprehensive advantages and prospects, CC technologies, combined with the utilization of lunar soil resources, can significantly improve construction efficiency and sustainability, reduce the transportation cost of Earth's materials, and are suitable for the construction of key facilities such as lunar landing platforms and residential cabins [72,73].

2.4. Summary

A summary of the features of 3D printing lunar regolith simulants formed by low-temperature deposition is presented in Table 1.

3. High-Energy Beam Additive Manufacturing

This process involves two distinct sintering/melting methods: powder bed fusion (PBF) and directed energy deposition (DED). The former involves the selective layer-wise melting of pre-spread regolith powder using a high-energy beam, whereas the latter directly applies concentrated energy to melt raw materials during simultaneous deposition.

3.1. Powder bed fusion (PBF)

3.1.1. Selective laser sintering/melting (SLS/SLM)

Experimental studies demonstrated the feasibility of SLM for fabricating lunar regolith simulant components, albeit with performance limitations inherent to the irregular morphology, thermal instability, and porosity of the material. The formed sample of the example is shown in Figs. 9(a–d), and the relevant process parameters and test results are listed in Table 2. As the results show, optimized energy densities (0.68–4 J/mm²) and adaptive scanning strategies (e.g., contour-infill speed differentiation, orthogonal hatching) achieved compressive strengths (4.2–31.4 MPa) and hardness values (657–1245 HV), making them potentially suitable for rudimentary structural applications, such as dust shields or pavement blocks.

Table 1
Summary of some key features in lunar regolith AM by low-temperature deposition forming.

Category	Materials	Process features	Formed sample features	Refs. No.
DIW (MEX)	Biopolymer bound	Regolith simulants: JSC-1A, LHS-1; Additive: human/ bovine/mammalian serum albumin, bovine blood proteins (AP920)	moisture-dependent curing;	uniaxial compressive strength: 25.0 MPa (no urea), 39.7 MPa (urea addition); limited thermal cyclability
	Matrix Transformation	Regolith simulants: lunar mare regolith simulant; Additive: hydroxyethyl cellulose	dissolution-precipitation; low energy consumption;	compressive strength: 6 MPa; density: 1.51~1.69 g/mL (before curing), 1.63~1.74 g/mL (after curing); liquid phase recycling; weak vacuum stability
	Alkali-activated	Regolith simulants: JSC-1A, BH-1, JLU-LRS, DNA-1, HIT-LRS-1; Additive: alkali activators	curing under vacuum environment;	compressive strength: 48.1 MPa; flexural Strength: 4.1 MPa; freeze-thaw resistant; anisotropic layer bonding
	Hydrogel systems	Regolith simulants: LHS-1, EAC-1A; Additive: hydrophilic polymer	dry in ambient conditions; require post-sintering; require intensive sintering energy	compressive strength: 0.68~2.4 MPa; porosity: 2.18~6.4 %; density: 0.78~0.89 g/cm ³ ; support complex geometries;
	Thermoplastic composites	Regolith simulants: artificial lunar soil (ALS); Additive: ABS	requires 235 °C extrusion temperature; temperature-sensitive processing	with the increase of ALS content, the toughness decrease, while the hardness and strength increase
FDM (MEX)	Thermoplastic composites	Regolith simulants: lunar regolith simulant; Additive: Nylon	requires powder pre-processing; rapid manufacturing	tensile strength: 36.2 MPa; impact toughness: 5.15 kJ/m ²
		Regolith simulants: LMS-1D; Additive: PEEK	filaments stored in dryer at 65 °C for 12 h; samples ironed at 5 mm/s with an extrusion flow rate of 10 %	density: 1.335~ 1.419 g/cm ³ ; tensile strength: 78.19 MPa at 12.25 % strain (15 wt % regolith), 67.05 MPa at 7.76 % elongation (30 wt % regolith)
		Regolith simulants: ore, BP-1, LHS-1, CLRS-1; Additive: PLA	processing temperatures from hopper to die ranged from 160 °C to 195 °C	flexural strength: 17.8 MPa (perpendicular to the grain), 33.4 MPa (parallel to the grain)
		Regolith simulants: MGS-1.; Additive: sulfur	rapid solidification (the 7-day strength was regarded as the ultimate strength)	compressive strength: 40.5~46.1 MPa; flexural strength: 7.1~7.9 MPa
	Sulfur concrete	concrete mixtures	large-scale construction; template-free rapid construction; low-gravity rheology issues	under evaluation
CC technologies			300-nozzle printhead; vacuum chamber operation; functional tool fabrication capability	
D-shape		inorganic binder mixtures		dimensions: 395 mm × 395 mm × 195 mm

Table 2
Summary of SLM-formed lunar regolith simulant samples: process parameters and performance results.

Laser power (W)	Scan speed (mm/s)	Energy density (J/mm ²)	Key parameters	Porosity (%)	Hardness (HV)	Compressive strength (MPa)	Elastic modulus (MPa)	Notes	Refs. No.
45	400 (contour), 190 (infill)	–	Adaptive scan strategies	–	1245	–	–	High geometric accuracy, smooth surfaces, superior hardness compared to fused silica	[74]
50	210	1.011	300 μm spot size, 35 % track overlap	40.8	670 ± 11	–	–	Optimal parameters for JSC-1A simulant; high visible-to-NIR absorbance (92 %)	[75]
–	320	0.68~0.92	Hatch spacing: 170~250 μm	44~49	657 ± 14	4.2 ± 0.1	287.3 ± 6.7	Strength comparable to masonry brick; failure behavior akin to ceramic foams	[76]
–	–	2.96~4	Highland simulant NU-LHT-2M	–	~695	~22.7	–	Demonstrated feasibility for structural applications; scalability potential noted	[77]
–	–	–	Refractory clay substrate	–	>680	>31.4	–	Optimal base plate material for multilayer stability; high yield stress	[78]
80	175	–	0.1 mm layer thickness	–	100~720	–	–	Severe brittleness observed (shattered upon impact); inhomogeneous microstructure	[79]
10~80	0.1~1.4	–	1.95 mm spot diameter, track overlapping ratio <44 %	–	–	–	–	Bubble encapsulation (43~67 % pore area) reduced density (1.65~2.40 g/cm ³); balling defects at high layer thickness (>2 mm)	[80]

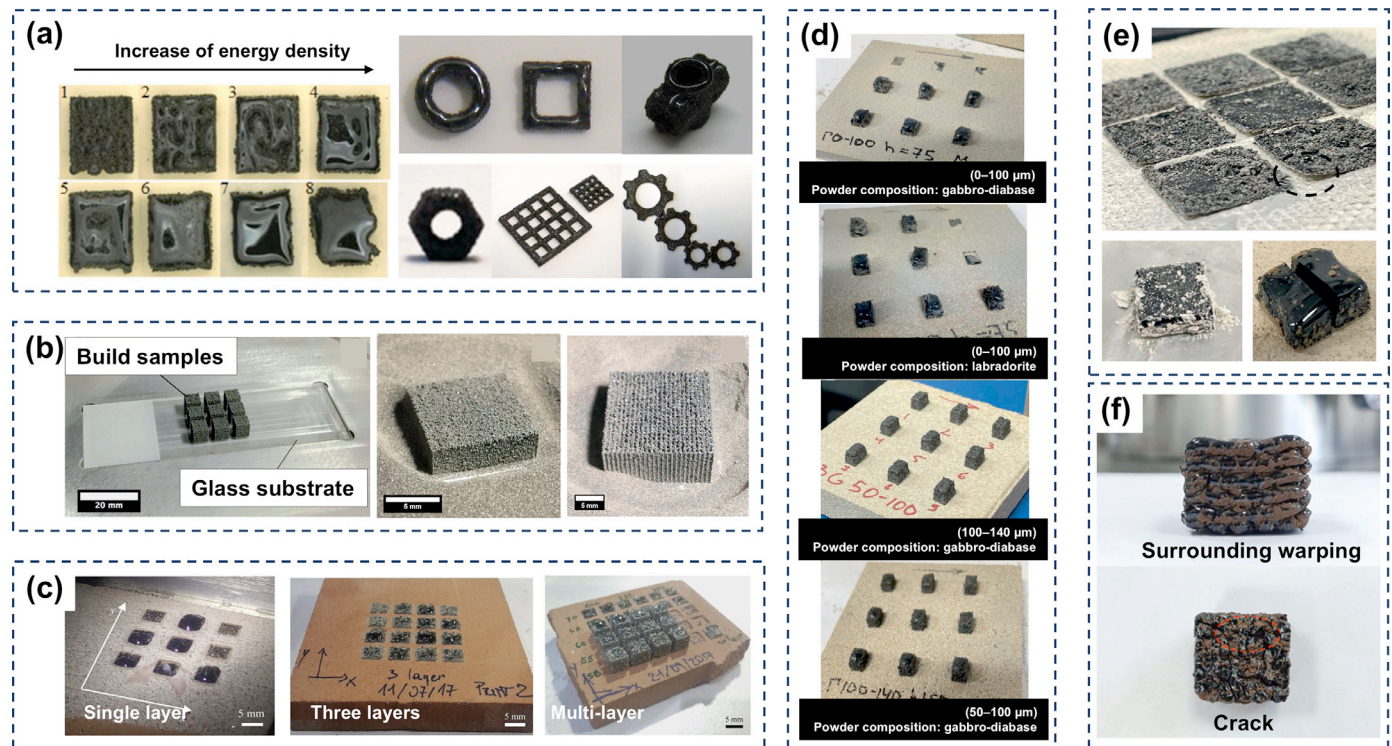


Fig. 9. Lunar regolith-based objects fabricated by SLM: (a) Multiple 3D objects [74,83]; (b) Cuboid samples [75,76,84]; (c) Single-layer and multilayer deposition [78]; (d) Fusion on four compositions on substrates [79]; (e) Multilayer fusion and fused sample [85]; (f) Defects in multilayer samples [81].

However, the residual porosity (40 %–49 %) and brittle fracture modes remain critical challenges linked to insufficient particle fusion and bubble encapsulation during rapid solidification (Fig. 9(e)). In multi-track fabrication, layer thickness studies identified a critical range of 3.16–3.82 mm (laser power-dependent), beyond which melt pool instability caused track fractures (Fig. 9(f)). Sub-2 mm layers improved surface quality through enhanced melt-substrate wetting but generated residual thermal stresses that promoted warping and microcracking [81,82].

A study employing hot-stage microscopy (HSM) on the JSC-2 regolith demonstrated sintering initiation at 1090 °C on platinum substrates, with material softening observed between 1130 °C and 1185 °C [86]. Optimal processing parameters were identified within the range of 1185–1250 °C, where contact angles decreased from 76° to 22°, significantly reducing layer defects such as delamination and balling. Enhanced bonding was achieved on the presintered substrates, with complete coalescence occurring at 1200 °C (Fig. 10(a)). Large laser spots (10–100 mm) under reduced pressure improved surface continuity [87], with crack formation linked to cooling gradients (Fig. 10(b)). The monitoring process using the high-speed camera is illustrated in Fig. 10(c). A 300 W diode laser revealed a 'wrapping mechanism' under large-beam, low-energy conditions, enabling continuous track formation through cyclic melting and melt retraction [88–90]. Melt pool stratification and a 60 W power threshold for stable tracks were identified in powder bed fusion [91], highlighting process–structure relationships for in-situ lunar manufacturing. Excessive energy induces keyhole-mode defects, whereas amorphous basalt compositions reduce the porosity by 30 %–40 % through viscous flow [92].

In pioneering experiments on lunar regolith simulant AM processing under microgravity ($\sim 10^{-6}$ g) and lunar gravity (0.16 g), adjustable gravity conditions and vacuum/atmospheric environments were accessible through the Einstein-Elevator (Fig. 11) [93].

This facility allows 4 s free-floating intervals for experiments, with rapid reset times of 4 min between trials. The key findings include the successful laser melting of regolith simulants under vacuum and Earth's

gravity (1 g), with subsequent validation under extra-terrestrial conditions. Computed tomography (CT) scans of the processed samples revealed round geometries, surface irregularities, and interconnected pores of varying sizes, highlighting the challenges in porosity control. The study emphasized the advantages of vacuum-based processing, such as reduced oxidation and impurity-free metal compounds, while underscoring the cost-effectiveness of the Einstein-Elevator for simulating space environments compared to orbital experiments. This study was supported by the MOONRISE payload installed on a mobile manipulator [94].

3.1.2. Selective solar light sintering (SSLS)

The Cassegrain Solar Concentrator System (Fig. 12), developed under NASA's in-situ resource utilization (ISRU) initiative, represents a critical advancement in extraterrestrial resource processing technologies [95]. This compact system, with a primary mirror diameter of 0.5 m, employs a dual-mirror configuration to concentrate solar energy for high-temperature applications, specifically targeting the melting of lunar regolith. The incident sunlight is reflected by the primary mirror onto a secondary mirror, which redirects the concentrated beam through an aperture at the base of the primary mirror, focusing behind the system.

The RegoLight project, led by the DLR and funded by the EU, developed solar sintering-based 3D printing for lunar infrastructure using regolith simulants [96–100]. Three printers (Ambient, Vacuum, Mobile) were designed for sintering under various conditions. A solar furnace with a 52 m² heliostat achieved 20 kW power and 4 MW/m² peak flux (Fig. 13(a)), while xenon lamps (1.2 MW/m² over Ø20mm) enabled sunlight-independent sintering [99] (Fig. 13(b)). Optimized parameters (48 mm/s scan speed, 14 mm spacing, 0.1 mm layer thickness) yielded layered structures, although the compressive strength remained at <5 MPa owing to interlayer bonding issues evidenced by X-ray tomography showing 0.6 mm open pores (Fig. 13(c)). A three-stage sintering approach was used (Fig. 13(d)): 0D point sintering (470 W/m² DNI, 4 s exposure), 1D line sintering (800 W/m², 10 mm/s scan speed), and 2D

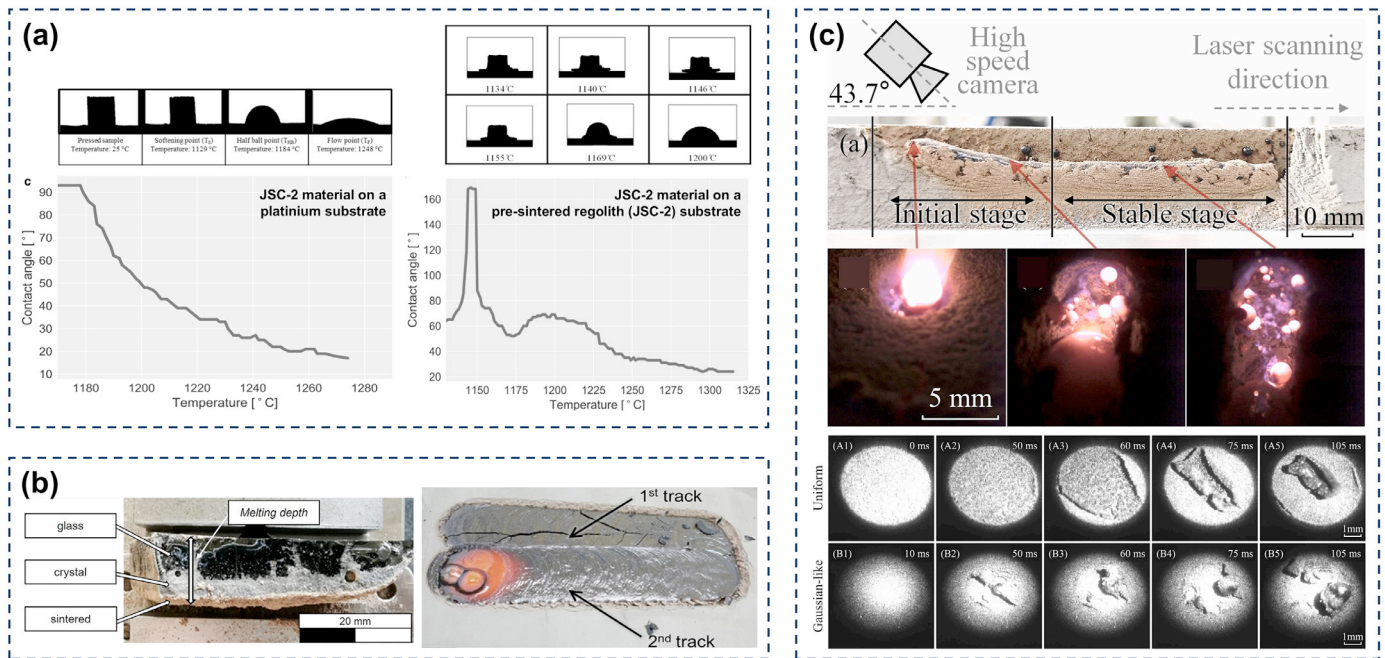


Fig. 10. (a) Measured effective contact angle versus temperature [86]; (b) Solid-liquid interfaces in laser fusion [87]; (c) High-speed camera monitoring for laser melting process showing “wrapping mechanism” [90].

Table 3
Mechanical properties of solar-sintered lunar regolith samples [97,98].

Type	Samples	Orientation	Average strength (MPa)	Average modulus (GPa)
Compressive strength	20	Perpendicular to layers	2.49 ± 0.71	0.21 ± 0.15
Flexural strength	30	Parallel to layers	0.55 ± 0.33	0.11 ± 0.08
	30	Perpendicular to layers	0.23 ± 0.10	0.03 ± 0.02

area sintering (16 mm inter-line spacing). In addition, sintering on a porous substrate helps to reduce warpage (Fig. 13(e)). Thermal stress-induced warping was mitigated through alternating scan patterns and the use of a porous substrate. In the practice of solar AM of lunar regolith [97,98], concentrated xenon lamps (12 mm diameter spot with 1200 kW/m^2 flux density) were used to sinter JSC-2A simulant layer-by-layer (100 μm thickness) at 48 mm/s, achieving structural components with anisotropic mechanical behavior, such as compressive strength and flexural properties (Fig. 13(f)). These results highlight the challenges in interlayer bonding homogeneity and defect sensitivity exacerbated by small sample geometries (10–20 mm).

The mechanical test results from the experiment are shown in Fig. 13(f) and listed in Table 3.

Optical waveguide (OW) is another choice for solar transmission, based on which researchers from Physical Science Inc. developed thermal power system for lunar ISRU applications [101–104]. In the solar thermal system, as shown in Fig. 14(a), solar radiation is collected by the concentrator and transmitted to the transmission line made of a low-loss optical fiber in the OW array. The system enabled heating regolith to 1800–1900 °C, enabling oxygen production via carbothermal reactions in practice [101,102]. Subsequent adaptations expanded its application to regolith sintering [103,104], producing a 15×15 sintered pad. As shown in Fig. 14(b), a solar concentrator power system was proposed to be carried on the athlete robot platform as part of a freeform additional construction system (FACS) [105,106].

Compared to other methods, the use of Fresnel lenses for solar concentration is the most widely used approach. Early studies by NASA teams [107,108] established foundational solar sintering principles us-

ing concentrated sunlight ($>1350 \text{ }^\circ\text{C}$) through mobile Fresnel lenses, achieving a sintering depth of 2.5 cm at $13 \text{ cm}^2/\text{min}$ while identifying critical challenges in beam tracking and deep-layer heating. This concept evolved into automated systems, such as Kayser’s solar sinter [109], which introduced computer-controlled X-Y-Z motion platforms and layered deposition techniques for glass formation at 1400–1600 °C. Fresnel lenses enable solar concentration to melt lunar dust, which has sufficient strength to produce an adhesive for printing structures [110]. Lunar-specific applications were performed by researchers from HIT [111] (Fig. 15), where an integrated solar powder bed fusion system featuring sun-tracking mechanisms (30° – 69.1° solar elevation adaptability), auxiliary heating (177 °C preheating), and high-energy-density focusing (3.36 – 3.87 MW/m^2).

Parametric studies established optimized operational thresholds [113]: optimized solar sintering parameters include $\geq 1200 \text{ W/m}^2$ irradiance for phase transition, a speed of 0.25 mm/s at 5 mm layers, and $\leq 1.5 \text{ mm}$ hatch spacing. Multilayer trials achieved $3 \text{ cm} \times 9 \text{ cm}$ structures with controlled defects, revealing bubble patterns and cooling effects. Key challenges remain regarding the scaling laws, subsurface microcracks, and uneven crystallization. Solar concentrators show ISRU potential, but require improved thermal management for scalable manufacturing.

3.1.3. Selective microwave sintering (SMWS)

Selective Microwave Sintering (SMWS) leverages the unique microwave-absorbing properties of lunar regolith to enable in-situ fabrication of structural and functional components [114]. A proposed “road-paving wagon” as shown in Fig. 16(a) employs dual magnetron arrays

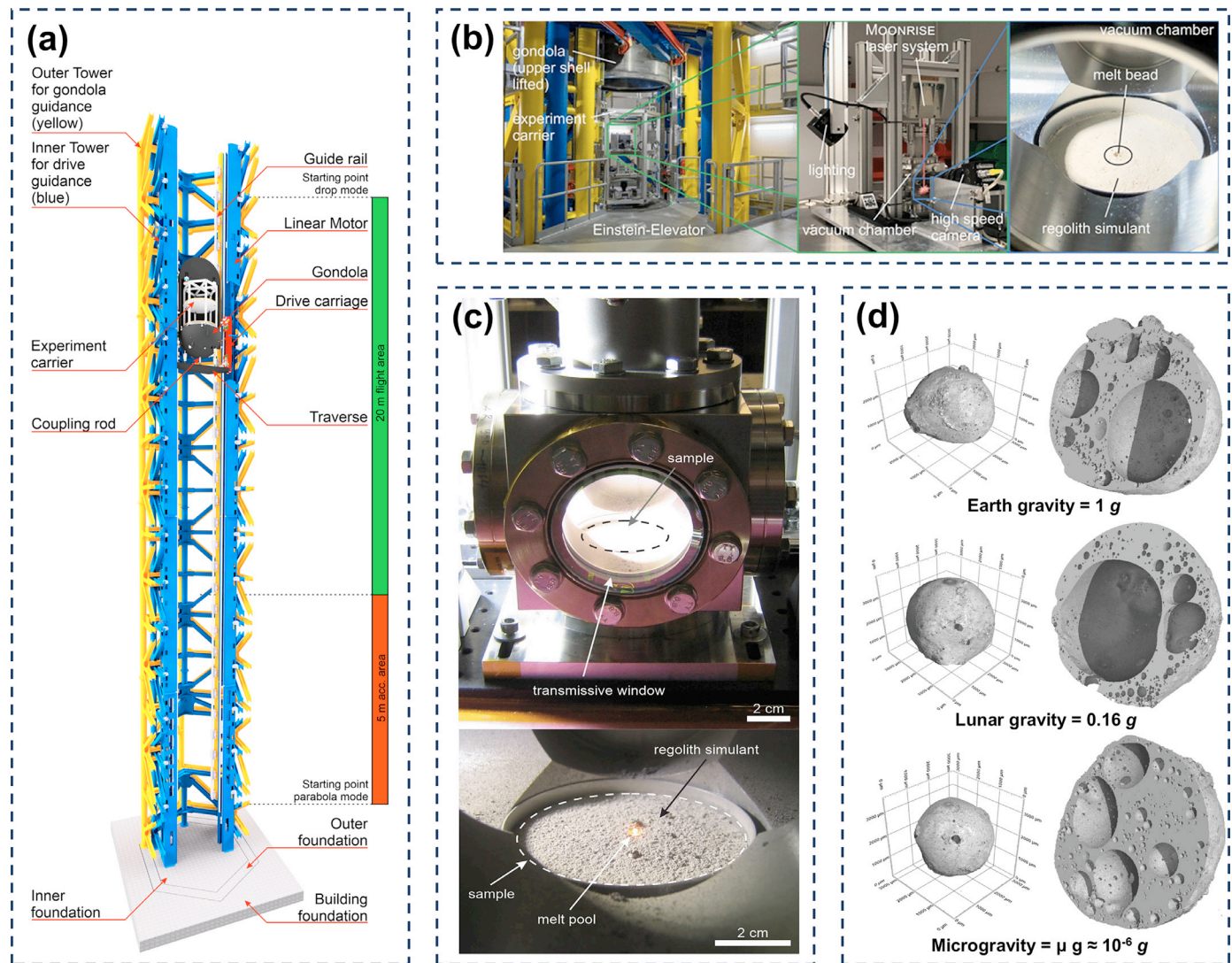


Fig. 11. AM processing under adjustable gravity conditions: (a, b) Diagram and setup of the Einstein-Elevator; (c) Mobile laser-melting vacuum chamber; (d) CT-scans of samples processed at various gravity conditions [93].

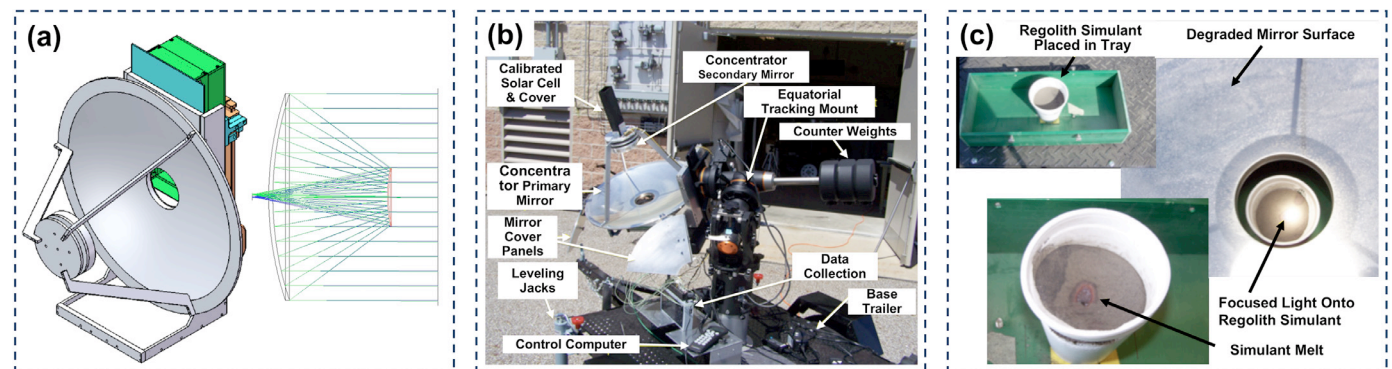


Fig. 12. Cassegrain Solar Concentrator System by NASA [95]: (a) Structural design layout of the ray trace; (b) Outdoor testing of the solar concentrator; (c) Initial melt test.

to sinter regolith to a depth of 0.5 m and melt the surface layer (3–5 cm) into glass, simultaneously releasing trapped solar-wind volatiles (e.g., H, He, C, N) for capture and utilization. The SMWS also supports the manufacture of prefabricated blocks for construction (e.g., igloos), forming parabolic antenna dishes via surface melting, and producing

glass fibers or wool from low-viscosity mare soils. To complement this, a free-form additive construction system (FACS) integrates microwave printheads for additive manufacturing [106], combining regolith sintering with concurrent volatile extraction during layer-by-layer deposition (Fig. 16(b)).

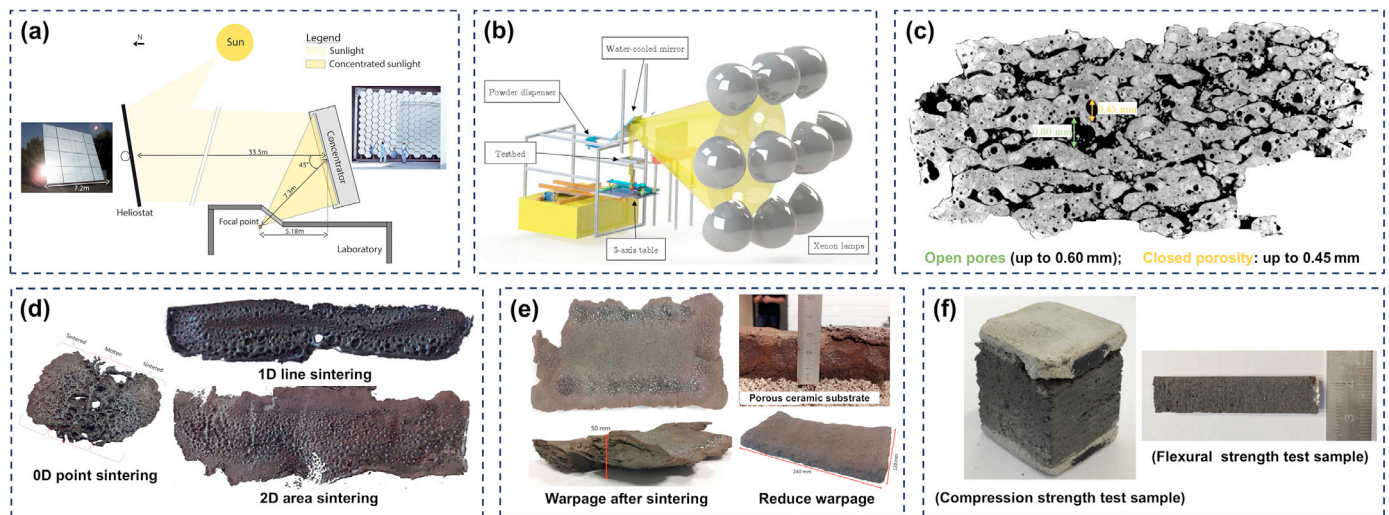


Fig. 13. (a) Solar oven; (b) Solar simulator; (c) X-ray tomography on samples; (d) Sintering in 0D, 1D and 2D; (e) Warpage reduction with porous substrate [99]; (f) Solar-sintered sample for mechanical tests [97,98].



Fig. 14. (a) OW solar thermal power system [101–104], and (b) mobile OW system mounted on ATHLETE [105,106].

3.2. Directed energy deposition (DED)

DED, or LENSTM (i.e., laser engineered net shaping from Optomec company), is an additive manufacturing process characterized by the simultaneous delivery of feedstock material (powder or wire) and high-energy input (typically a laser or electron beam) to achieve localized melting and controlled deposition. Balla et al. [115] demonstrated the feasibility of fabricating dense, near-net-shaped lunar regolith simulant structures using a 0.5 kW continuous-wave Nd-YAG laser on an aluminum alloy substrate. Optimal parameters (50 W laser power, 20 mm/s scan speed, and 12.36 g/min powder feed rate) yielded defect-free cylindrical parts (Fig. 17). A calculated energy density of 2.12 J/mm ensured adequate melt-pool formation for interlayer bonding (30 % remelting of

the prior layers) while mitigating issues such as excessive spreading or cracking.

Apart from the abovementioned techniques, Vat Photopolymerization, particularly stereolithography (SLA) and digital light processing (DLP), also demonstrates significant potential for lunar regolith additive manufacturing. SLA achieved a compressive strength of 444 MPa with 62 vol % regolith paste [116], while DLP attained a flexural strength of 91 MPa via silane-modified slurries (60 wt %–65 wt %) [117–119]. Sintering optimization achieves 95 % relative density in highland variants [120], with TiO₂ content (2.1 wt %–8.7 wt %) critically influencing densification [121,122]. DLP demonstrates material adaptability through particle engineering [123], although SLA outperforms DLP in terms of strength (444 vs. 222–428 MPa). Recent advances have established the

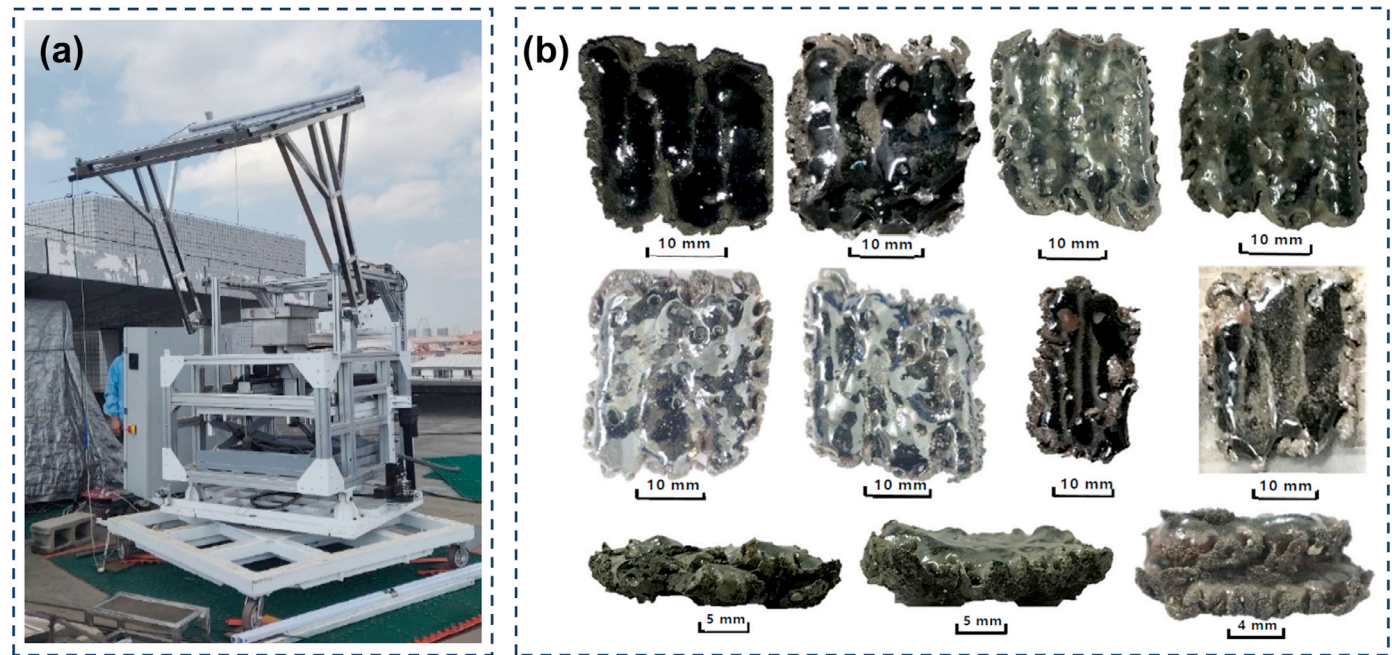


Fig. 15. (a) Lunar regolith solar additive manufacturing testing system, and (b) printed products by HIT [111,112].

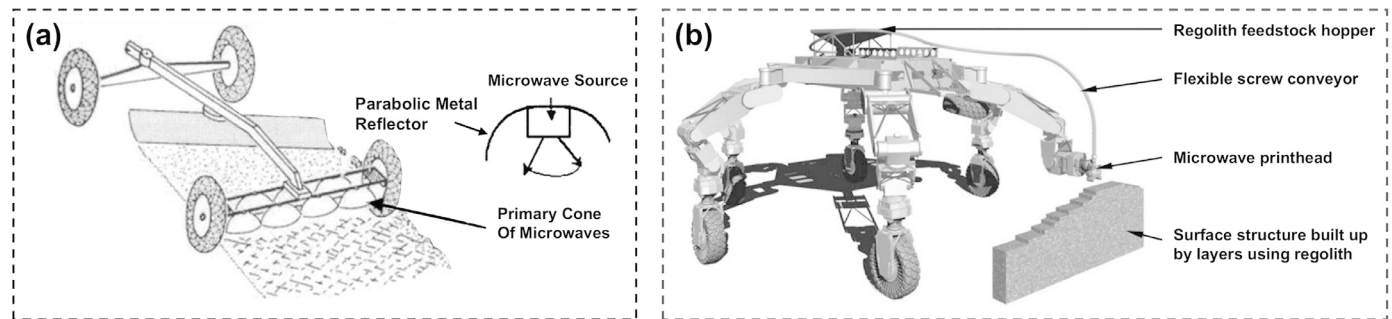


Fig. 16. Selective microwave sintering of lunar regolith: (a) “Lunar Road-Paving Wagon” [114], and (b) microwave printhead on Freeform Additive Construction System (FACS) [106].

Table 4
Summary of some key features in lunar regolith AM by high-energy beam additive manufacturing.

Category		Materials	Process features	Formed sample features	Refs. No.
Powder bed fusion (PBF)	Selective laser sintering/melting (SLS/SLM)	Regolith simulants: JSC-1, JSC-1A, JSC-2, JSC-2A, LHS-1, FJS-1, EAC-1A, NU-LHT-2 M, CUG-1A, HIT-I-1	laser power: 10W~11kW scan speed: 0.1~300 mm/s energy density: 0.68~4 J/mm ² layer thickness: 0.1~2 mm laser spot size: 15μm~6.95 mm	compressive strength: 17~31.4 MPa; porosity: 30~49 %; hardness: 100~1245 HV brittle fracture	[74–94]
	Selective solar light sintering (SSLS)	Regolith simulants: JSC-1, JSC-1A, JSC-2A, CUG-1A;	solar power: 0.09~387 W/cm ² ; solar spot size: 12~21.8 mm layer thickness: 0.1 mm; scanning speed: 10~47 mm/s	compressive strength: 2.31~2.49 MPa, compressive Young's modulus: 0.14~0.21 GPa flexural strength: 0.55 MPa (parallel to the layer's plane), 0.23 MPa (perpendicularly to the layer's plane); flexural Young's modulus: 0.11 GPa (parallel), 0.03 GPa (perpendicularly)	[95–113]
	Selective micro-wave sintering (SMWS)	Regolith simulants: MLS-1, JSC-1	sintering depth: 0.5 m; surface melting depth: 3~5 cm;	glassy surface layer formation; porous structures; unique for volatile extraction during sintering	[106,114]
Directed energy deposition (DED)		Regolith simulants: JSC-1A	laser power: 50 W; scan speed: 20 mm/s; powder feed rate: 12.36 g/min; energy density: 2.12 J/mm ² ; spot size: 1.65 mm;	dense, crack-free structures; dense structure with precise interlayer remelting; limited to small-scale components	[115–124]



Fig. 17. Solid cylindrical lunar regolith simulant parts made using LENS™ [115].

thermal stability (500 °C) and hardness (~9.3 GPa) of DLP ceramics [124], emphasizing the need for integrated material process databases to advance lunar manufacturing.

3.3. Summary

A summary of the features of 3D printing lunar regolith simulants produced by high-energy beam additive manufacturing is presented in Table 4.

4. Analysis and Discussion

The AM of lunar regolith encompasses two primary technological paradigms: low-temperature deposition forming and high-energy beam additive manufacturing. These approaches differ fundamentally in their

energy utilization, material requirements, and consolidation mechanisms, presenting unique trade-offs for extra-terrestrial construction. AM technologies have been explored for processing lunar regoliths. Table 5 provides a comparative analysis of current applications and future prospects.

The following analysis synthesizes engineering feasibility across four critical dimensions—energy demand, transportation logistics, process complexity, and structural performance—to evaluate their suitability for lunar infrastructure development.

(1) Energy requirements

The energy availability on the Moon remains a pivotal constraint. Low-temperature deposition-forming technologies, such as MEX and BJT, operate at lower energy thresholds during the printing phase. MEX leverages ambient or solar-assisted curing for geopolymers or sulfur concrete, consuming 0.1–0.5 kWh/kg. However, post-processing steps—such as thermal curing (60–100 °C for geopolymers) or UV-assisted crosslinking—increase energy demands. BJT typically achieves structural consolidation through ambient-phase binder solidification, whereas certain BJT subtypes employing binders may utilize optional post-sintering for enhanced mechanical performance, introducing additional energy inputs that create a tradeoff between energy efficiency and material performance across BJT variants. By contrast, high-energy beam additive manufacturing technologies require concentrated energy inputs. Laser-based PBF consumes 3–5 kWh/kg owing to high melting temperatures (1100–1600 °C), while solar sintering (SSLS) relies on intermittent lunar daylight, achieving flux densities of 1.2–4 MW/m². Solar systems face efficiency losses owing to dust accumulation and beam-tracking challenges, particularly during the Moon's two-week night period. Although energy intensive, DED offers localized melting with reduced material waste, making it suitable for small-scale repair.

(2) Transportation logistics

Payload mass restrictions prioritize technologies that minimize Earth-derived inputs. Low-temperature deposition-forming methods depend on binders such as polymers, sulfur, or alkali activators (e.g.,

Table 5
Comparative analysis of lunar regolith AM techniques.

Category		Current status	Advantage	Challenges	Future prospects
Low-temperature Deposition Forming	Material extrusion (MEX)	Energy consumption: Low (ambient/solar drying); Comprehensive Mechanical Strength: 9~48 MPa; ISRU Compatibility: Moderate (binder-dependent)	Versatile feedstock (polymers, geopolymers); Low energy consumption	Binder dependency reduces ISRU efficiency; Layer adhesion issues in vacuum	Hybrid systems combining geo-polymerization and ISRU-derived binders; optimization for vacuum/low-gravity extrusion
	Binder jetting (BJT)	Energy consumption: Low (ambient/solar drying); Comprehensive Mechanical Strength: 2~5 MPa (post-sintered); ISRU Compatibility: Low (binder-dependent)	High scalability; Minimal energy input; Compatible with large structures	Low mechanical strength (2~5 MPa); Dependency on binders; Layer warping	Optimization of binder chemistry and sintering parameters for vacuum conditions; integration with robotic systems for habitat construction
High-energy Beam AM	Powder bed fusion (PBF)	Energy consumption: High (laser/solar); Comprehensive Mechanical Strength: 2.31~31.4 MPa; ISRU Compatibility: High	High ISRU compatibility (no binders); Solar energy utilization	Low mechanical integrity (solar sintering); High porosity (40~49 %); Slow process	Scalable solar concentrators; multi-energy hybrid systems (laser + solar); improved layer bonding strategies
	Directed energy deposition (DED)	Energy consumption: High (laser); Comprehensive Mechanical Strength: 8~15 MPa; ISRU Compatibility: High	High-density outputs; Low material waste; Suitable for repairs	High energy demand; Limited geometric complexity; Equipment portability	Development of portable energy systems; application in critical component repair or small-scale fabrication

NaOH), which constitute 5 %–30 % of the feedstock by mass. For instance, urea-modified geopolymer formulations have demonstrated enhanced workability with 3 wt %–5 wt % urea additive, while making it unable to get rid of terrestrial additive dependencies. BJT implementations using specialized binders (e.g., organic binders) present unique ISRU challenges. By contrast, high-energy beam additive manufacturing systems, particularly PBF and DED, eliminate binder dependency by using raw regolith directly. However, they require the transportation of heavy equipment such as laser systems, solar concentrators (0.5–1 m diameter mirrors), or microwave arrays. Recent advances in lightweight diode lasers and deployable solar reflectors have improved their feasibility; however, their scalability remains limited by launch mass budgets.

(3) Process complexity

Operational simplicity is critical for the deployment of autonomous robots. MEX involves multistep workflows such as regolith–binder mixing, extrusion, and curing. Vacuum conditions destabilize water-based binders (e.g., hydrogels), whereas thermal cycling (−190 °C to 127 °C) induces layer delamination in geopolymers. A BJT requires precise binder–powder interactions to mitigate warping, compounded by the Moon's low gravity (1/6 g, or 1.62 m/s²), which reduces powder-bed density and stability. However, high-energy beam additive manufacturing processes face distinct challenges. Laser PBF struggles with irregular particle morphology of the regolith, leading to incomplete fusion and porosity. The broad beam profile of solar sintering (~20 mm spot size) limits the resolution, whereas vacuum conditions exacerbate thermal stress cracking. DED's simultaneous deposition and melting of DED enables rapid fabrication but poses uncontrollable risks under vacuum, restricting applications to simple geometries such as landing pads.

(4) Structural performance

The mechanical robustness under lunar conditions dictates the viability of the technology. Low-temperature deposition forming-produced geopolymers achieved the highest compressive strength (48 MPa) but exhibited anisotropic layer bonding, reducing the flexural strength by 20 %–30 %. Sulfur concrete, with rapid curing (12 h to 80 % strength), suffers from vacuum-induced micropores and long-term sulfate degradation. The BJT-sintered parts remained brittle (<5 MPa) and were suitable only for non-load-bearing elements. Meanwhile, high-energy-beam additive manufacturing outputs exhibit superior homogeneity but limited toughness. The laser-sintered regolith reaches a compressive strength of 31 MPa but fractures catastrophically under impact owing to residual porosity. Solar-sintered components exhibit even lower strengths (2–5 MPa) owing to poor interlayer fusion. There has not been much research on the DED method, and the fabricated samples have serious cracking problems that are not yet usable.

(5) Regolith properties

Authentic lunar regolith exhibits sharp interlocking particles owing to space weathering, in contrast to the engineered spherical morphologies of terrestrial simulants optimized for flowability. This divergence critically impacts AM compatibility: low-temperature deposition processes (e.g., MEX and BJT) prioritize binder adhesion and particle packing, favoring controlled gradation of simulants but masking challenges from the abrasive angularity of the regolith. Conversely, high-energy beam processes (e.g., PBF and DED) rely on material-energy interactions dictated by the native properties of the regolith, including heterogeneous mineralogy affecting melt dynamics, nanophase iron altering radiation absorption, and irregular morphology impeding powder bed uniformity. Although simulants enable terrestrial process validation, their homogenized chemistry/sphericity risks underestimating authentic regolith processing complexities, necessitating lunar-specific adaptations in both binder-based and energy-driven AM paradigms.

5. Conclusion and Prospects

This review systematically evaluates lunar-regolith-based AM technologies and highlights their potential for enabling sustainable extraterrestrial construction. Low-temperature deposition methods excel in energy efficiency and material versatility but face ISRU compatibility limitations, while high-energy beam additive manufacturing offers binder-free fabrication at the cost of energy intensity and geometric constraints. To address the "5W1H" challenges (Why, When, Where, Who, What, How) of lunar construction, the following research priorities are proposed:

(1) Multi-scale characterization of lunar regolith engineering properties

Future efforts must prioritize resolving the technical bottlenecks in the multiscale mechanical characterization of lunar regoliths. Microscale studies should focus on in-situ particle morphology, gradation distribution, and force transmission mechanisms, whereas macroscale investigations should address the rheological behavior, bearing capacity, and stability under simulated lunar conditions. These insights will enable the precise optimization of 3D printing process parameters.

(2) In-situ material formation and property regulation

The solidification mechanisms of lunar regolith in extreme lunar environments (vacuum, microgravity, and thermal cycling) require urgent clarification. Research should establish morphology–property relationships for regolith-based materials with varying mineral compositions and particle sizes, enabling the targeted regulation of material performance through compositional design and process optimization.

(3) Integrated structure–material–process co-design

A holistic design framework must be developed to address synergistic challenges in the lunar environment, including thermal stress, micrometeoroid impacts, and radiation exposure. This approach should optimize the structural efficiency of large-scale habitats while achieving multifunctional requirements, such as radiation shielding, thermal insulation, and seismic resilience, through hierarchical material architectures and adaptive manufacturing strategies.

(4) Autonomous robotic construction systems

The transition from proof-of-concept prototypes to operational systems requires breakthroughs in lightweight reconfigurable 3D printing platforms capable of autonomous operation under lunar conditions. Key priorities focus on developing algorithms for multi-robot collaboration, integrating vision-force hybrid sensing for real-time quality control, and creating construction management systems that combine human, machine, and environmental factors—supported by digital twin technologies.

(5) High-fidelity lunar environment simulation

To bridge the gap between laboratory research and mission-ready technologies, advanced ground-based test beds must be created to simulate the Moon's extreme conditions. These platforms should replicate reduced gravity, high vacuum, thermal extremes, and cosmic radiation simultaneously, enabling full-scale validation of construction processes and systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wucheng Sun: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Fen Dang:** Validation, Investigation. **Yan Zhou:** Writing – review & editing, Supervision. **Shifeng Wen:** Validation, Resources. **Cheng Zhou:** Supervision, Resources, Project administration. **Yusheng Shi:** Supervision, Resources. **Lieyun Ding:** Supervision, Funding acquisition.

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